

Project: Multi-VLAN Network Design for a University IT Department

Scenario

Designed and configured a secure multi-VLAN network for a university IT department using Cisco Packet Tracer. The network separated different departments into individual VLANs to improve security, reduce broadcast traffic, and simplify network management. Implemented Router-on-a-Stick (ROAS) for inter-VLAN routing and developed a complete IP subnetting plan to efficiently allocate addresses for each department. Added also DHCP to enable dynamic IP distribution.

Network Requirements

- Faculty Department – 45 devices
- Computer Laboratory – 90 devices
- Administration Office – 20 devices
- Server Room – 10 devices
- Network Management VLAN – 5 devices

Subnetting Plan

VLAN	Department	Required Hosts	Subnet	Mask	Gateway
VLAN 10	Faculty	45	192.168.10.0/26	255.255.255.192	192.168.10.1
VLAN 20	Laboratory	90	192.168.20.0/25	255.255.255.128	192.168.20.1
VLAN 30	Administration	20	192.168.30.0/27	255.255.255.224	192.168.30.1
VLAN 40	Server Room	10	192.168.40.0/28	255.255.255.240	192.168.40.1
VLAN 99	Management	5	192.168.99.0/29	255.255.255.248	192.168.99.1

Devices Used

- 1 Cisco 2911 Router
- 3 Cisco 2960 Switches
- PCs for each department
- Servers